

Grievance Handling System

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1. Project Title

- Grievance Handling System (GHS)

2. Objective

- To build a simple desktop application where employees can register their workplace complaints digitally.
- To make sure all complaints are stored safely in a database so management can track, manage, and resolve them without losing any data.

3. Technologies Used

- Backend Language: Java (OOP concepts like Classes, Objects, and Lists).
- Database Platform: MySQL (To store users and grievance records securely).
- Database Driver: JDBC (Using PreparedStatement and ResultSet to connect Java with MySQL).
- Development Environment: Eclipse IDE and Git/GitHub for version control.

4. Problem Solved by the Project

- No More Lost Complaints: In many offices, people complain via verbal talks or random emails, which easily get forgotten or ignored. This project puts everything into a central system.
- No Tracking Guesswork: Before this system, employees never knew if anyone was working on their issue. Now, they can see live status updates.
- Better Management Oversight: Managers and admins don't have to search through paper files; they can view a clean list of all open and closed issues instantly.

5. Advantages

- **Simple & Organized:** It uses clean menus that anyone can navigate.
- **Safe Storage:** Because it uses MySQL, data is safe even if the Java application is closed or restarted.
- **Privacy & Security:** Employees cannot access management tools, and managers cannot accidentally delete records.

6. Features

- **Three Dynamic User Roles:** The system changes its options automatically based on who logs in (Admin, Employee, or Manager).
- **Live Status Changes:** Every complaint starts as "Pending" and changes to "Resolved" once a manager fixes it.
- **Crash Protection:** The menus use String inputs, meaning the application won't crash if a user accidentally types a letter instead of a number.

Role Tiers	Main Capabilities & Actions
System Admin (Role ID: 1)	Can view the entire System Audit Log or permanently Delete a grievance record to clean up the database.
Employee (Role ID: 2)	Can Register/Login , Raise a Grievance (with a subject and details), and Search for existing tickets using keywords.
Grievance Manager (Role ID: 3)	Can View All Reports in the system and change a ticket status to " Resolved ".

8. SQL Features (Special Features in Your Code)

- **Fuzzy Keyword Search:** The search feature doesn't require an exact match. It uses the SQL LIKE operator with wildcards (%keyword%), meaning searching for "fan" will instantly find "fan broken" or "ceiling fan is not working".
- **Built-in Validation Rules:** The system checks data before sending it to the database. For example, if an employee tries to submit a ticket with a title shorter than 3 letters, the system blocks it and shows an error message.
- **Clean Architecture Design:** The codebase is split into professional design layers:
 - **Model Tiers (Grievance.java, User.java):** Holds the raw data structures.
 - **DAO Tiers (GrievanceDAO.java, UserDAO.java):** Handles raw SQL queries (INSERT, SELECT, UPDATE, DELETE).
 - **Service Tiers:** Manages business rules and input validation.
 - **Controller Tier (GrievanceController.java):** Manages the text menus and reads inputs from the user.